Table 1: CUDAHeat — Enrichment Rules Catalogue

| Long name | Formula | Description | Interpretation |
|---|---|---|---|
| $F_T$ ([-])<br>**Threshold exceedance** – Fraction of event hours above the climatological threshold<br>**Cat**: persistence, threshold exceedance | $$F_T = \frac{\sum_{t=t_s}^{t_e} \mathbf{1}\left[SST_{p,t} > T_{p,h(t)}\right] \mathbf{1}_{\text{valid}}(p,t)}{\sum_{t=t_s}^{t_e} \mathbf{1}_{\text{valid}}(p,t)}$$ | Fraction of valid hourly observations within a detected marine heatwave event for which SST exceeds the corresponding climatological percentile threshold. | Values approaching 1 indicate persistent threshold exceedance throughout the event. Lower values indicate a larger contribution from below-threshold periods, such as internal gaps or joined event segments. |
| $DSEF$ ([-])<br>**Daytime strong exceedance fraction** – Fraction of daytime event hours exceeding 1.5 times the climatological threshold anomaly<br>**Cat**: diurnal cycle, daytime intensity | $$DSEF = \frac{\sum_{t=t_s}^{t_e} \mathbf{1}_{\text{day}}(p,t)\mathbf{1}\left[SST_{p,t} > C_{p,h(t)} + 1.5\left(T_{p,h(t)} - C_{p,h(t)}\right)\right] \mathbf{1}_{\text{valid}}(p,t)}{\sum_{t=t_s}^{t_e} \mathbf{1}_{\text{day}}(p,t)\mathbf{1}_{\text{valid}}(p,t)}$$ | Fraction of valid daytime hourly SST observations during a marine heatwave event exceeding 1.5 times the difference between the climatological threshold and climatological mean. Daytime is defined as 06:00–20:00 local solar time, calculated from UTC timestamps and longitude. | Values approaching 1 indicate frequent strong daytime thermal exceedances. Values approaching 0 indicate that daytime SST rarely reaches 1.5 times the climatological threshold anomaly. This metric measures the frequency of strong daytime warming rather than its absolute magnitude. |
| $HC$ ([°C h])<br>**Heat Content** – Cumulative thermal exceedance above the climatological threshold<br>**Cat**: intensity, cumulative exceedance | $$HC = \sum_{t=t_s}^{t_e} \max\left(SST_{p,t} - T_{p,h(t)}, 0\right) \Delta t$$ | Cumulative positive SST anomaly above the climatological 90th-percentile threshold over the entire marine heatwave event. | Higher values indicate greater accumulated thermal intensity. Hours below threshold contribute zero, including internal gaps and periods joining separate exceedance segments. |
| $NBCF$ ([-])<br>**Nocturnal below-climatology fraction** – Fraction of nighttime event hours below the climatological mean<br>**Cat**: diurnal cycle, nocturnal recovery | $$NBCF = \frac{\sum_{t=t_s}^{t_e} \mathbf{1}_{\text{night}}(p,t)\mathbf{1}[SST_{p,t} < C_{p,h(t)}]\mathbf{1}_{\text{valid}}(p,t)}{\sum_{t=t_s}^{t_e} \mathbf{1}_{\text{night}}(p,t)\mathbf{1}_{\text{valid}}(p,t)}$$ | Fraction of valid nighttime hourly SST observations during a marine heatwave event that fall below the corresponding hourly climatological seasonal mean. Nighttime is defined as 20:00–06:00 local solar time, calculated from UTC timestamps and longitude. | Values approaching 1 indicate frequent nighttime SST conditions cooler than the climatological seasonal mean. Values approaching 0 indicate that nighttime SST generally remains above normal climatological conditions. This metric measures the frequency of nocturnal recovery below climatology, not the magnitude of cooling. |
| $NBTF$ ([-])<br>**Nocturnal below-threshold fraction** – Fraction of nighttime event hours below the climatological threshold<br>**Cat**: diurnal cycle, nocturnal recovery | $$NBTF = \frac{\sum_{t=t_s}^{t_e} \mathbf{1}_{\text{night}}(p,t)\mathbf{1}[SST_{p,t} < T_{p,h(t)}]\mathbf{1}_{\text{valid}}(p,t)}{\sum_{t=t_s}^{t_e} \mathbf{1}_{\text{night}}(p,t)\mathbf{1}_{\text{valid}}(p,t)}$$ | Fraction of valid nighttime hourly SST observations during a marine heatwave event that fall below the corresponding climatological percentile threshold. Nighttime is defined as 20:00–06:00 local solar time, calculated from UTC timestamps and longitude. | Values approaching 1 indicate frequent nighttime interruption of marine heatwave conditions. Values approaching 0 indicate persistent nighttime threshold exceedance. This metric measures the frequency of below-threshold conditions, not the magnitude of cooling. |
| $NDHR$ ([-])<br>**Nocturnal Daytime Heat Retention** – Fraction of nights exceeding the preceding daytime maximum threshold<br>**Cat**: diurnal cycle, nocturnal heat retention | $$NDHR = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}\left[\min_{t\in\mathcal{N}_n} SST_t > \max_{t\in\mathcal{D}_n} T_t\right]$$ | Measures the fraction of complete nights during which SST remains continuously above the maximum climatological threshold of the preceding daytime period. Daytime is defined as 06:00–20:00 local solar time. Nighttime is defined as 20:00–06:00 local solar time. For each night, the reference is the maximum hourly climatological threshold observed during the preceding day. A night qualifies only if every nighttime SST observation strictly exceeds this daytime maximum threshold. | Values approaching 1 indicate that SST frequently remains above the preceding daytime maximum threshold throughout the entire night. Values approaching 0 indicate that nocturnal SST frequently falls below the preceding daytime maximum threshold. A value of 1 indicates that all complete valid nights satisfy the thermal retention condition. This metric measures sustained nocturnal thermal intensity relative to the preceding daytime threshold, rather than ordinary nighttime threshold exceedance. |